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Ferrous Cast Ductile Iron Piston Introduction

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This service/parts topic is to introduce the new single piece Ferrous Cast Ductile (FCD) iron pistons for the QSK45 and QSK60 engines and secondly to introduce a new Service Practice change for articulated pistons used in QSK45 and QSK60 engines.

The articulated pistons used in the QSK45 and QSK60 have been obsoleted and superseded to new single piece ferrous cast ductile iron pistons. These new QSK45/60 ferrous cast ductile pistons are same technology pistons used in the QST30 engine.

The ferrous cast ductile piston **must** be used in combination with a new hardened cylinder liner, piston ring set, and piston pin. The new cylinder liner has been nitride hardened for improved robustness and supersedes the old cylinder liner. The new cylinder liner is dimensionally the same as its predecessor. The new top piston ring is a cast iron ring with a wrap around hard particle chrome material and its width is narrower than its articulated piston's counterpart. The piston ring gap is also narrower. The piston pin is unique to the ferrous cast ductile piston. The ferrous cast ductile piston pin is similar in appearance to the K38/50 piston with square cut ends as compared to the articulated piston pin's concave ends.

When the new hardened cylinder liner is unused it is duller in appearance as compared to the old cylinder liner because of the nitride hardening. However, the only certain method to tell the difference between the cylinder liners is by part number identification, stamped into the cylinder liner. See Fig. 1. Visual inspection of the top piston ring is also required to determine correct piston ring part number. The top piston ring **must** be installed correctly, top side up, because the ring face is no longer symmetric. See Fig. 2.

Note : The ferrous cast ductile piston was designed to have the same weight as the articulated piston. As a result, both piston types can be intermixed in an engine in any number combination and regardless of engine bank side.

Note : The component part numbers listed in Tables 1, 3 and 5 are for reference only. Refer to the Service Kits Part Numbers in Tables 2, 4, and 6 to order Cummins Service Kits.

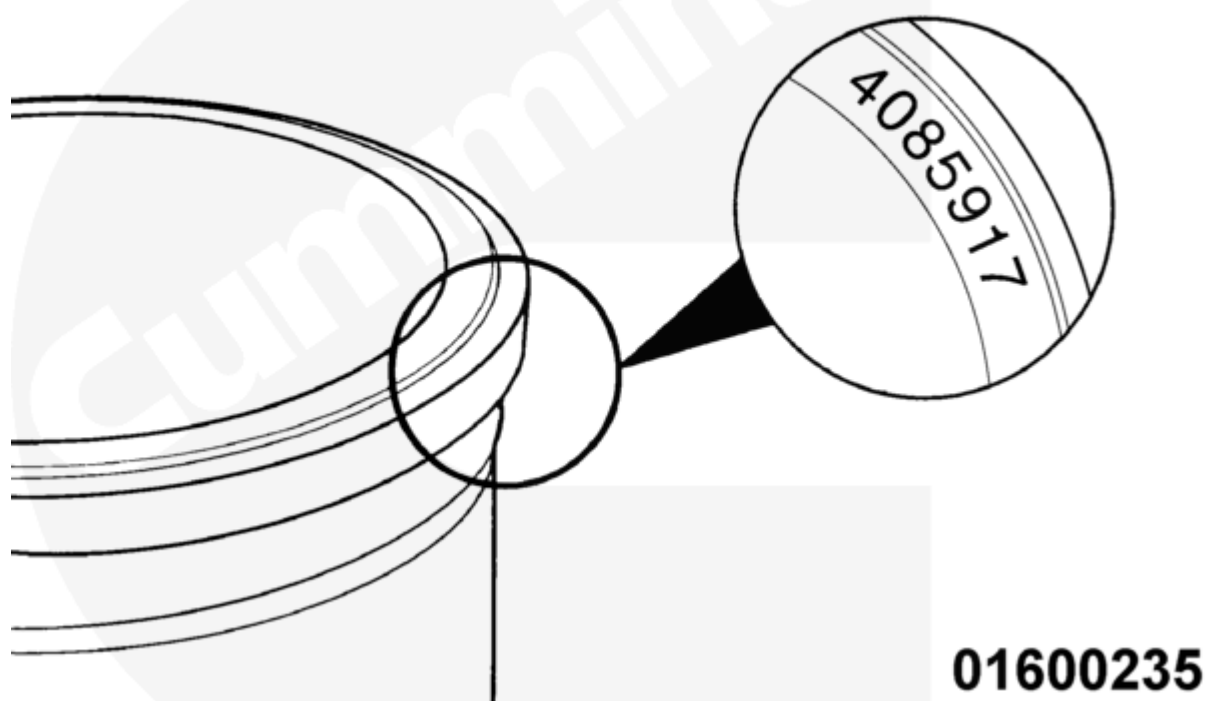


Fig. 1. Cylinder Liner Part Number Location

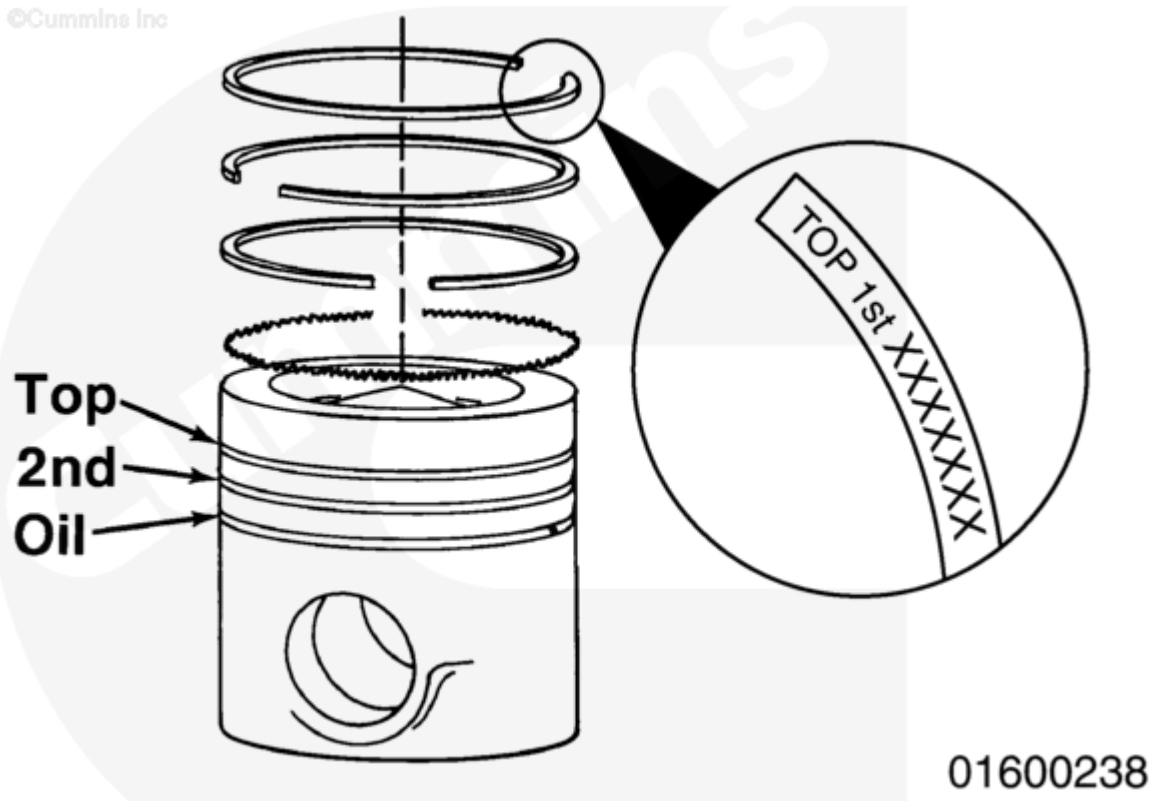


Fig. 2. Piston Ring Part Number Location

Table 1: High Brake Mean Effective Pressure (BMEP) Piston Component Part Numbers			
Component Description	Old Part Number for Articulated Power Cylinder	New Part Number for FCD Power Cylinder	Remarks
Piston	4017319	4066543	Old part number is top crown of articulated piston
Piston Skirt	3411262	No Supersession	Skirt is obsolete.
Piston Pin	3089013	4001416	Unique FCD piston pin
Top Piston Ring	3070589 or 3410861	4100481	Narrower ring for FCD piston. No supersession on 3070589 since it used on the QSK19.
2 nd Piston Ring	3090460	3090460	No Change
Oil Piston Ring	3090450	3090450	No Change
Snap Ring	3089251	3089251	No Change
Cylinder Liner	3410342	4085917	Hardened liner required for FCD piston.

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Table 2: High BMEP Piston Service Kits			
Service Kit Description	Old Part Number	New Part Number	Remarks
Piston Kit	4001105	4089246	
Cylinder Liner Kit	4024811	4089143	Hardened Liner
Piston Ring Kit	3631248 or 4089145	4089247	Narrower Top Ring for FCD Piston
CPL No.			2666, 2667, 2820, 2821, 2824, 2848, 2849, 2850, 2851, 2888, 2920, 8173, 2655, 2656, 2665, 2823, 2854, 2935, 8156

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Table 3: Low BMEP Piston Component Part Numbers			
Component Description	Old Part Number for Articulated Power Cylinder	New Part Number for FCD Power Cylinder	Remarks
Piston	4017521	4066551	Old part number is top crown of articulated piston
Piston Skirt	3411262	No Supersession	Skirt is obsolete.
Piston Pin	3089013	4001416	Unique FCD piston pin
Top Piston Ring	3070589 or 3410861	4100481	Narrower ring for FCD piston.
2 nd Piston Ring	3090460	3090460	No Change
Oil Piston Ring	3090450	3090450	No Change
Snap Ring	3089251	3089251	No Change
Cylinder Liner	3410342	4085917	Hardened liner required for FCD piston.

Table 4: Low BMEP Piston Service Kits			
Service Kit Description	Old Part Number	New Part Number	Remarks
Piston Kit	4001107	4089268	
Cylinder Liner Kit	4024811	4089143	Hardened Liner
Piston Ring Kit	3631248 or 4089145	4089247	Narrower Top Ring for FCD Piston
CPL No.			2855, 2948, 2469, 2853, 8155

Table 5: TA Luft (European Emissions) Piston Component Part Numbers			
Component Description	Old Part Number for Articulated Power Cylinder	New Part Number for FCD Power Cylinder	Remarks
Piston	4065430	4067004	Old part number is top crown of articulated piston
Piston Skirt	3411262	No Supersession	Skirt is obsolete.
Piston Pin	3089013	4001416	Unique FCD piston pin

Table 5: TA Luft (European Emissions) Piston Component Part Numbers			
Component Description	Old Part Number for Articulated Power Cylinder	New Part Number for FCD Power Cylinder	Remarks
Top Piston Ring	3070589 or 3410861	4100481	Narrower ring for FCD piston.
2 nd Piston Ring	3090460	3090460	No Change
Oil Piston Ring	3090450	3090450	No Change
Snap Ring	3089251	3089251	No Change
Cylinder Liner	3410342	4085917	Hardened liner required for FCD piston.

Table 6: TA Luft Service Kits			
Service Kit Description	Old Part Number	New Part Number	Remarks
Piston Kit	Not Applicable	4089269	New FCD Kit
Cylinder Liner Kit	4024811	4089143	Hardened Liner
Piston Ring Kit	3631248 or 4089145	4089247	Narrower Top Ring for FCD Piston
CPL No.			2869, 2819

Table 7: Compatibility Matrix				
Compatibility Matrix	New FCD Piston	New FCD Piston Top Ring	Hardened Cylinder Liner	New FCD Piston Pin
Articulated Piston (Crown and Skirt)	Intermix in engine allowed with appropriate cylinder liner, piston ring and pin combination.	NO	YES	NO
Articulated Piston Ring Set	NO	Not Applicable	Only Service Kit Part Number 4089145	Not Applicable
Old Cylinder Liner (Soft)	NO	NO	Not Applicable	NO
Articulated Piston Pin	NO	Not Applicable	Not Applicable	Not Applicable

These product changes will be phased into production at the Daventry Engine Plant and Cummins India Ltd, starting in April 2002 (ESN 1st33151884) for the High BMEP piston and July 2002 for the Low BMEP piston.

Note : This is a product improvement, not subject to campaign or temporary repair practice.

The Service Practice change covered in this Service Parts Topic is to **not** reuse the articulated piston crown nor the articulated piston skirt when an articulated piston is removed. Discard the articulated piston (crown and skirt) power cylinder and install a ferrous cast ductile power cylinder that includes ferrous cast ductile piston, ferrous cast ductile piston ring set, ferrous cast ductile piston pin and hardened liner. For engines inside of base warranty, standard warranty guidelines apply.

Note : This is a product improvement, not subject to campaign or temporary repair practice.

The ferrous cast ductile piston can be reused when the power cylinder is removed if it checked OK using the following inspection procedure.